

4 Electrical Measurements - Site Distribution Board

4.1 Residual Current Devices (RCD) Testing

Verification of RCD parameters: trip current (I_{Δ}) and trip time (t).

No.	Circuit Name	U_n	I_n	$I_{\Delta n}$	I_{Δ}	t [ms]			Status
		[V]	[A]	[mA]	[mA]	$1 \times I_{\Delta n}$	$2 \times I_{\Delta n}$	$5 \times I_{\Delta n}$	
1	AIRCONDITION L1-F6 C20	230.00	20.00	30.00	21.60	179.00	83.00	38.00	✓ OK
2	L1-F7 C16 Socket1	230.00	16.00	30.00	25.80	16.00	6.00	5.00	✓ OK
3	L1-F8 C16 Socket12	230.00	16.00	30.00	27.20	17.00	6.00	5.00	✓ OK
4	L1-F9 C16 Rack1	230.00	16.00	30.00	-	-	-	-	⊖ NO ACTION
5	L1-F10 C16 Rack2	230.00	16.00	30.00	25.80	16.00	5.00	5.00	✓ OK

4.2 Fault Loop Impedance & Shock Protection

Verification of automatic disconnection of supply (Z - impedance, I_k - short circuit current).

No.	Circuit Name	I_n [A]	Z_s [Ω]	I_k [A]	Status
1	USV-L1-PE-F5-C50	50.00	0.30	775.00	✓ OK
2	USV-L2-PE-F5-C50	50.00	0.28	808.00	✓ OK
3	USV-L3-PE-F5-C50	50.00	0.29	785.00	✓ OK
4	AIRCONDITION L1-F6 C20	20.00	0.33	696.00	✓ OK
5	L1-F7 C16 Socket1	16.00	0.33	691.00	✓ OK
6	L1-F8 C16 Socket12	16.00	0.28	817.00	✓ OK
7	L1-F9 C16 Rack1	16.00	0.32	715.00	✓ OK
8	L1-F10 C16 Rack2	16.00	0.33	707.00	✓ OK
9	L1-F11-C20	20.00	0.27	865.00	✓ OK

4.3 Earth Resistance Measurements

Measurements performed according to EN 62305 standard using the technical method.

No.	Measurement Point	R_s [Ω]	R_a [Ω]	Status
1	Front	7.60	20.00	✓ OK
2	Rear	7.56	20.00	✓ OK

Weather conditions during measurement: SUNNY

4.4 Continuity of Earthing & Main Connections

Verification of protective bonding and continuity for all metallic enclosures and infrastructure elements.

Circuit	Cable type	Status	Circuit	Cable type	Status
Cabinet 1	LGY 16mm2	✓ OK	Cabinet 2	LGY 16mm2	✓ OK
ODF1	LGY 16mm2	✓ OK	ODF2	LGY 16mm2	✓ OK
USV	LGY 16mm2	✓ OK	DUCTS	LGY 6mm2	✓ OK
Grounding strip	LGY 50mm2	✓ OK	Klima	LGY 16mm2	✓ OK
Floor	LGY 16mm2	✓ OK	Power Distribution	LGY 25 mm2	✓ OK